Project Initialization and Planning Phase

Project: OptiCrop - Smart Agricultural Production Optimization Engine

Section	Content
Objective	Develop an AI-powered crop recommendation system using soil nutrients and weather data.
Scope	Recommend suitable crops to improve productivity and support smart agriculture.
Problem	Farmers face difficulty selecting crops due to poor soil analysis and changing climate.
Impact	Improves crop yield and decision making.
Approach	Use Machine Learning with Flask deployment.
Key Features	Crop recommendation, soil analysis, weather integration, fertilizer suggestions.

Resources

Resource	Details
Hardware	Intel i5, 8GB RAM, SSD
Framework	Flask
Libraries	scikit-learn, pandas, numpy, matplotlib
IDE	VS Code / PyCharm
Dataset	Crop Recommendation CSV, Soil & Weather Data